

sxFusion: a novel single-cell analysis tool based on cell feature fusion and co-optimization of low-rank representation with clustering

Abstract—Single-cell RNA sequencing (scRNA-seq) enables cellular-level gene expression analysis, but high sparsity and noise in the data pose significant challenges for accurate cell-type identification. Here, we propose sxFusion, a structure-aware clustering method that integrates low-rank representation (LRR) with orthogonal non-negative matrix factorization (NMF) to improve robustness. sxFusion introduces three main innovations: first, it integrates graph-based features from a GCN and nonlinear attribute embeddings from an MLP via a dimension-wise gating mechanism, enhancing feature discriminability and mitigating over-smoothing; second, it employs a structure-guided low-rank self-expression model jointly optimized with orthogonal NMF, aligning representation learning directly with clustering assignments and avoiding two-step inconsistencies; third, it outputs cluster labels along with a cluster allocation matrix, enabling downstream interpretation without additional clustering steps. Benchmarked against eleven state-of-the-art methods on eleven single-cell datasets, sxFusion consistently achieved higher NMI and ARI. Ablation and UMAP analyses confirmed the advantages of the gating-based fusion and joint optimization strategies. These results demonstrate that sxFusion effectively identifies biologically meaningful cell populations and marker genes, providing robust and interpretable clusters beneficial for pathway analysis, elucidation of disease mechanisms, and clinical stratification.

Keywords—GCN, gating mechanism, LRR, NMF, single cell clustering

I. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) technology has advanced our understanding of cellular heterogeneity by providing high-resolution gene expression profiles [1]. However, the considerable sparsity and technical noise inherent in scRNA-seq data present notable challenges for accurate cell type identification [2]. Although traditional clustering methods provide a foundation for single-cell data analysis, their performance may be affected by high dimensionality, matrix sparsity, and the presence of technical noise, which can obscure biological signals [3].

Recent computational methods have adopted a range of strategies to address these challenges. Graph-based approaches, such as GNMF [4], MPSSC [5], and Seurat [6], leverage cell-to-cell relationships to improve clustering performance. Methods based on low rank representation (LRR) [7], including KGLRR [8], and DLNLR [9], improve performance through dimensionality reduction and structure recovery. Neural-network-based approaches, such as sigDGCN [10], scGAC [11], and scGNN [12], further advance clustering performance by capturing nonlinear and hierarchical structures within the data. However, most of these methods perform clustering in two separate stages: first extracting features, then applying a

clustering algorithm. This separation may introduce mismatch between representation learning and clustering objectives, potentially reducing the accuracy of clustering [13]. Recently, DRjCC [14] has made progress by jointly optimizing dimensionality reduction and convex clustering within a unified framework, thus alleviating the inconsistency between feature learning and clustering. Nevertheless, DRjCC mainly focuses on feature embeddings and global structure recovery, but does not fully exploit local graph topology or integrate attribute and structure information at a deeper level.

Inspired by prior work, we propose the sxFusion method, a unified framework integrating structure-aware representation learning and clustering, as shown in Fig. 1. Specifically, sxFusion leverages a learnable gating mechanism to fuse two complementary views—topological features from a graph convolutional network (GCN) and nonlinear attribute embeddings from a multilayer perceptron (MLP)—thereby mitigating over-smoothing and preserving node-specific patterns. From the fused embeddings, sxFusion constructs a similarity graph that captures cell-to-cell affinities and guides a constrained low-rank self-expression model, enhancing robustness against dropout and noise. Unlike conventional two-step methods, sxFusion jointly optimizes LRR and orthogonal non-negative matrix factorization (NMF) [15], aligning representation learning with clustering assignments in a single step. Ultimately, sxFusion outputs stable and biologically interpretable clusters alongside a basis matrix and cluster allocation matrix, facilitating downstream analysis of heterogeneous single-cell datasets.

II. METHOD

A. GCN

Graph Convolutional Networks (GCNs) [16] are neural network models specifically designed for learning representations from graph-structured data. By recursively aggregating information from neighboring nodes, GCNs effectively integrate node attributes with graph topology. Specifically, GCN utilizes a symmetrically normalized adjacency matrix to perform weighted averaging and linear transformations of node features. Formally, the computation at l layer is defined as follows:

$$H^{(l)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l-1)} W^{(l)}), \quad (1)$$

where $H^{(l)}$ is the node feature matrix at the l -th layer, with $H^{(0)} = X$ as the input feature matrix. $\sigma(\cdot)$ denotes the activation function such as ReLU. $\tilde{A} = A + I$ is the adjacency matrix with self-loops. $\tilde{D} = \text{diag}(\tilde{A})$ is the degree matrix of $\tilde{A}$. $W^{(l)}$ is the trainable weight matrix at the l -th layer.

B. Gating fusion Mechanism

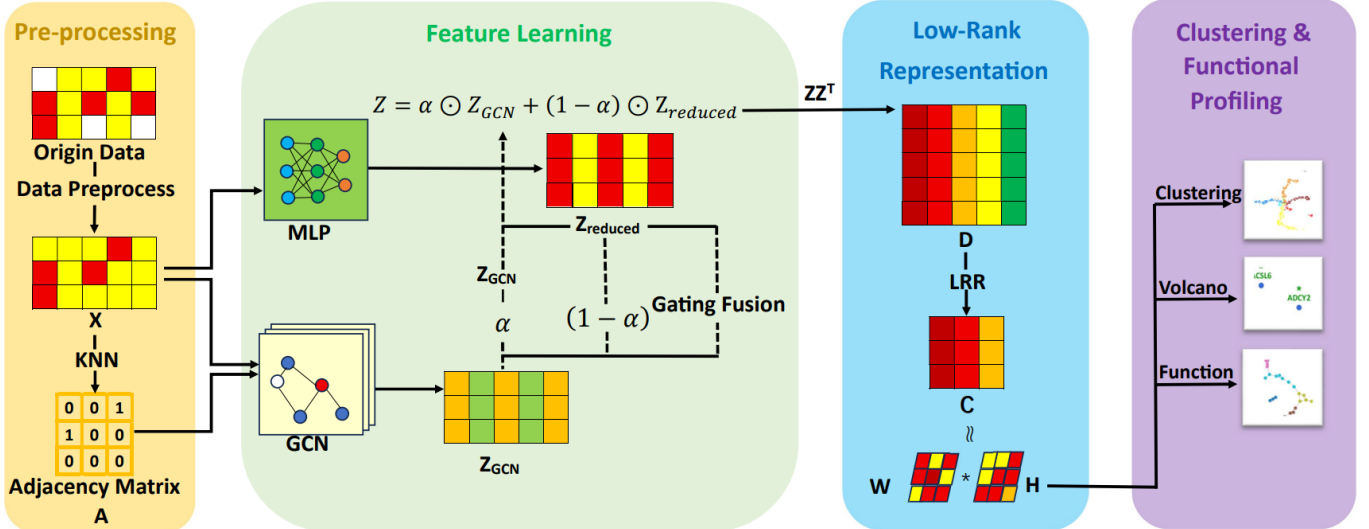


Fig. 1 The framework of sxFusion. (1) Gene filtering and L_2 norm are used to process the original samples. Gene filtering removes cells with non-zero gene expression levels of less than 5. (2) sxFusion first employs an MLP to extract and preserve the original attribute information of each sample, while a GCN is used in parallel to learn structural embedding features. A gating fusion mechanism then adaptively integrates the attribute and structural information, producing a unified representation that captures both aspects. Building on this fused representation, a LRR module models global sample-to-sample similarities, and a NMF module performs clustering assignments. This unified process yields both the cluster allocation matrix H and the final cluster labels. (3) The cluster allocation matrix H is

Gating fusion is a feature integration technique designed to adaptively balance the contributions from multiple feature sources or branches. Unlike simple concatenation or averaging, it introduces a learnable gating vector that controls the contribution of each input feature in the final fused representation. This enables fine-grained, data-dependent feature selection at each dimension. Mathematically, gating fusion is formulated as:

$$h_{fuse} = \alpha \odot h_1 + (1 - \alpha) \odot h_2, \quad (2)$$

where $h_1, h_2 \in R^d$ are the input feature vectors from two different sources such as a GCN branch and an attribute branch. $\alpha \in [0, 1]^d$ is the learnable gating vector, often computed as $\alpha = \sigma(MLP([h_1, h_2]))$, where σ denotes the sigmoid activation function and MLP is a multi-layer perceptron. $\odot$ denotes element-wise (Hadamard) multiplication. h_{fuse} is the fused feature representation. Gating mechanism is particularly effective for integrating heterogeneous or noisy features, as it adaptively suppresses uninformative or noisy components and emphasizes informative ones based on the input data. By doing so, it reduces the negative impact of feature heterogeneity or noise, leading to more robust and expressive fused representations [17].

C. Low-rank representation

The low-rank representation (LRR) [7] model captures global data structure by expressing each sample as a linear combination of all others. It seeks a low-rank coefficient matrix to represent data in a self-expressive form:

$$\min_C \|X - XC\|_F^2 + \lambda \|C\|_*, \quad (3)$$

$$\text{s.t. } C \geq 0, \text{diag}(C) = 0,$$

here, $X \in R^{n \times d}$ is the sample matrix, X each row represents a sample and each column a feature. $\|\cdot\|_F$ denotes the Frobenius norm. $C \in R^{n \times n}$ is the low-rank coefficient matrix. $C \geq$

$0, \text{diag}(C) = 0$ enforce non-negativity and prevent trivial self-representation. The nuclear norm $\|\cdot\|_*$ serves as a convex relaxation of the rank function, promoting low-rankness. λ is a regularization parameter.

D. NMF

Non-negative Matrix Factorization (NMF) [15] is an unsupervised learning technique that factorizes a non-negative sample matrix $X \in R^{n \times d}$ into two non-negative matrices for dimensional reduction or clustering:

$$\min_{W, H} \|X^T - WH\|_F^2, \text{ s.t. } W \geq 0, H \geq 0, \quad (4)$$

where $W \in R^{d \times k}$ contains k basis vectors. $H \in R^{k \times n}$ is the cluster assignment matrix, where each column of H encodes the soft membership of a sample in the k latent components.

E. sxFusion

Traditional LRR directly uses the raw sample matrix $X \in R^{n \times d}$ as a dictionary, which may make it vulnerable to noise or missing values [9]. In view of this, we incorporate a graph embedding matrix D derived from fused node features into the LRR framework. The embedding matrix is defined as:

$$D = ZZ^T, \quad (5)$$

here $Z = \alpha \odot Z_{GCN} + (1 - \alpha) \odot Z_{reduced}$ represents the fused node representations obtained via a gating mechanism, and $Z \in R^{n \times d_z}$. $\alpha \in [0, 1]^{n \times d_z}$ is a learnable gating vector. Z_{GCN} encodes structural features learned from GCN. $Z_{reduced} = f_{MLP}(X) \in R^{n \times d_z}$ is the output of a lightweight MLP applied to the original node attributes. This fusion mechanism adaptively integrates structural and attribute information. The LRR objective with graph embedding is formulated as:

$$\min_C \|X^T - X^T DC\|_F^2 + \lambda \|C\|_F^2, \quad (6)$$

$$\text{s.t. } C \geq 0, \text{diag}(C) = 0,$$

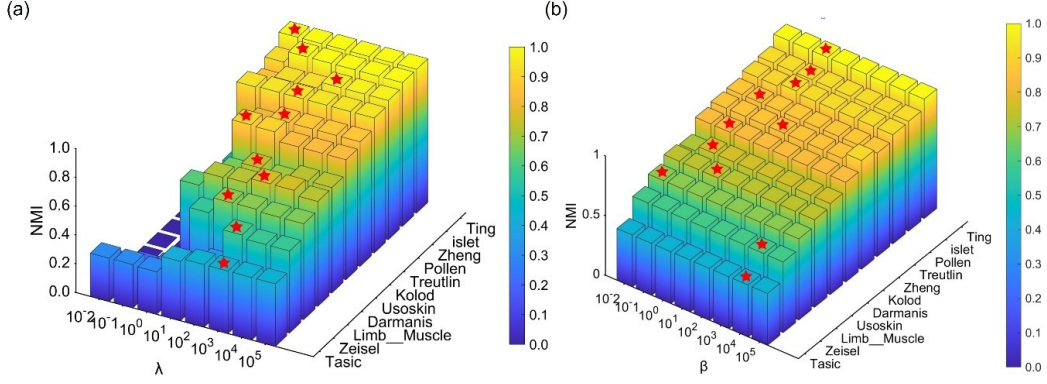


Fig. 2 Performance of the sxFusion with different values of λ and β . The bar plots show the NMI vales achieved on each dataset under different regularization strengths. The x-axes represent the values of λ and β , and the y-axis represents the NMI. Each group along the back x-axis corresponds to a different single-cell dataset. Red stars indicate the best NMI achieved for each dataset.

where $C \in R^{n \times n}$ is the self-expression coefficient matrix encoding inter-sample relationships, λ is the regularization parameter. The matrix D captures structural similarities among nodes, guiding the construction of a more robust LRR.

To avoid the inconsistency between feature learning and clustering often seen in two-step methods, sxFusion introduces a unified framework that jointly learns the LRR and cluster assignments. The overall objective is:

$$\min_{C, H, W} \|X^T - X^T A C\|_F^2 + \lambda \|C\|_F^2 + \beta \|C - WH\|_F^2, \quad (7)$$

$$\text{s.t. } C \geq 0, W \geq 0, H \geq 0, H^T H = I, \text{diag}(C) = 0,$$

where $W \in R^{n \times k}$ and $H \in R^{k \times n}$ are the basis and cluster allocation matrix of the NMF, respectively. $H^T H = I$ encourages distinct cluster memberships by enforcing decorrelation among cluster assignment vectors. β are regularization parameters. The framework of sxFusion is shown in Fig. 1.

III. RESULT AND ANALYSIS

A. Experimental setting

To assess the clustering performance of sxFusion on eleven single-cell datasets, we adopt normalized mutual information (NMI) [18] and adjusted rand index (ARI) [19] as evaluation metrics. Dataset details are provided in TABLE I. To balance the contributions of the coefficient matrix C and the clustering objective, sxFusion introduces two hyperparameters, $\lambda \in$

$[10^{-2}, 10^5]$ and $\beta \in [10^{-2}, 10^5]$. A grid search strategy is employed to identify the optimal parameter combination, as illustrated in Fig. 2.

sxFusion uses a two-layer GCN, and the default number of KNN neighbors is set to 5. Optimization is performed with Adam, using a learning rate of 10^{-2} and a regularization coefficient of 10^{-3} . The NMF module is run with a random seed of 42 for 200 iterations to ensure convergence and reproducibility. For general applications to other datasets, we recommend setting λ to 10^{-2} to 10^1 , and tuning β within the range $[10^{-2}, 10^0]$ for robust performance.

B. Performance

This section evaluates the performance of sxFusion on several scRNA-seq datasets, as shown in Fig. 3. All results reported are averaged over 50 independent runs of cluster allocation matrix. Baseline methods are implemented strictly following the official protocols and recommended settings from their original publications to ensure fair comparison.

On eleven scRNA-seq datasets, as presented in TABLE I, sxFusion achieves perfect clustering performance (NMI = 1.000, ARI = 1.000) on simpler datasets like Li_islet and Ting. On the more challenging Kolod dataset—characterized by moderate dropout (45.6%) and highly similar cell types—sxFusion achieves NMI of 0.946 and ARI of 0.961, improving on KGLRR by +12% (NMI) and +9.1% (ARI), and on Monocle3 by +4.5% and +12.3%, respectively. These improvements highlight the effectiveness of gating-based fusion of graph and attribute features and joint low-rank and NMF optimisation in capturing both local coherence and global structure.

Under extreme sparsity (Zheng, 88.76% dropout) sxFusion attains NMI = 0.980 and ARI = 0.988, surpassing DLNLR by +7.7% / +5.7% and DRjCC by +1.4% / +1.1%. This robust recovery stems from its structure-aware low-rank factorisation on fused embeddings.

Across scale (Limb Muscle, 3 909 cells, 47.7% dropout) sxFusion delivers NMI = 0.721 and ARI = 0.571. This represents +17.7% (NMI) / +3.8% (ARI) over KGLRR. This may be because sxFusion employs an adaptive gating fusion mechanism and a joint optimization strategy. These designs better integrate structural and attribute information and enhance robustness under high dropout and heterogeneity.

TABLE I ELEVN SCRNA-SEQ DATASETS.

Datasets	Genes	Cells	Dropout rate (%)	Classes	Accession
Li islet [22]	4494	60	0.00	6	GSE73727
Treutlein [23]	959	80	51.65	5	GSE52583
Ting [24]	14450	114	52.17	5	GSE51372
Pollen [25]	14805	249	50.79	11	SRP041736
Darmanis [26]	22085	420	81.40	8	GSE67835
Zheng [27]	4770	500	88.76	3	SRP073767
Usoskin [28]	17772	622	78.10	4	GSE59739
Kolod [29]	10685	704	27.87	3	E-MTAB-2600
Tasic [30]	5832	1727	32.71	48	GSE71585
Zeisel [31]	4412	3005	46.01	9	GSE60361
Limb Muscle [32]	1248	3909	47.74	6	GSE109774

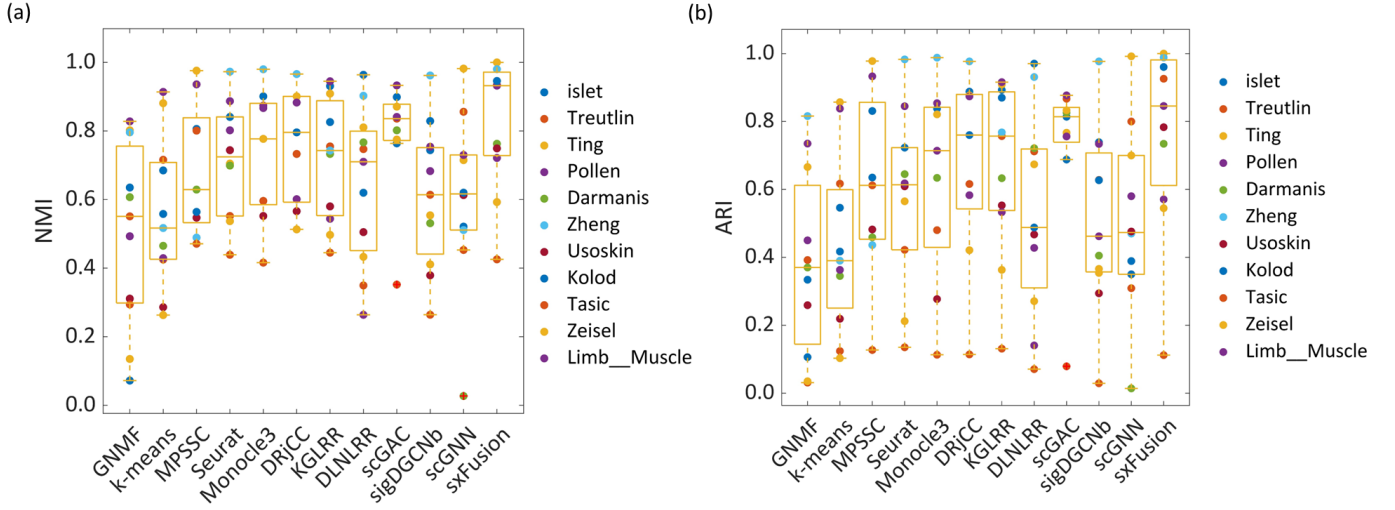


Fig. 3 Different methods are used to compare the clustering performance of 11 single-cell RNA-seq datasets on NMI and ARI.

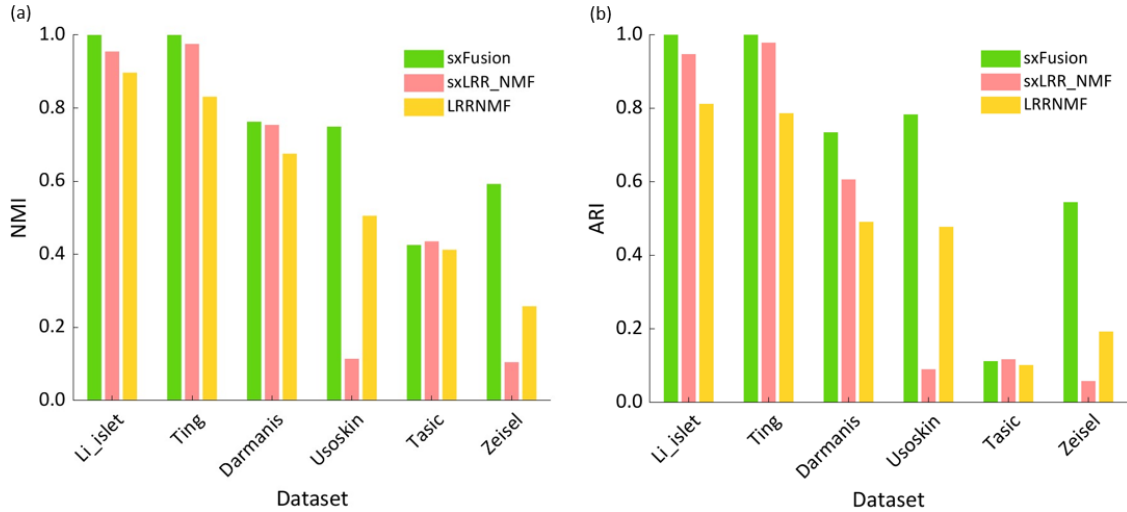


Fig. 4 Ablation results on six single-cell datasets. Left: NMI; right: ARI. Methods: sxFusion; sxLRR_NMF, which decouples NMF from LRR while retaining the GCN and the gating fusion mechanism; LRRNMF, which keeps only the joint LRR-NMF module and removes.

Collectively, these results show that while sxFusion’s unified framework—combining adaptive structure-aware fusion with clustering-driven low-rank optimisation—yields accurate, robust, and generally scalable clustering across a wide range of sparsity levels and dataset sizes, there remains room for improvement on the most complex scenarios where graph-based methods currently excel.

C. Effect of Model Components

We evaluate sxFusion and two ablated variants across six benchmark scRNA-seq datasets, as shown in Fig. 4. sxLRR_NMF removes joint optimization and performs NMF on the LRR derived from fused GCN and attribute features. LRRNMF preserves the joint LRR-NMF clustering mechanism but removes the GCN and gating fusion module, relying only on raw attribute features.

sxFusion consistently outperforms both baselines. It achieves the highest possible clustering scores (NMI = 1.000,

ARI = 1.000) on Li_islet and Ting. It also demonstrates strong performance on Darmanis (NMI = 0.762, ARI = 0.734) and Usoskin (NMI = 0.749, ARI = 0.783). Specifically, on Usoskin, sxFusion achieves an ARI of 0.783, which is nearly 9 times higher than that of sxLRR_NMF—highlighting the effectiveness of joint optimization in aligning representation learning with clustering objectives under complex structural patterns.

On sparsity datasets such as Tasic (32.71% dropout, 48 classes), where all methods exhibit limited performance, sxFusion nonetheless remains competitive. For instance, on Tasic, sxFusion achieves NMI/ARI of 0.426/0.112, which is only marginally lower than sxLRR_NMF’s 0.436/0.117, indicating comparable performance despite the challenging nature of the dataset.

These findings confirm that both gating-based structural fusion and joint optimization of representation and clustering are essential. Removing either component leads to substantial

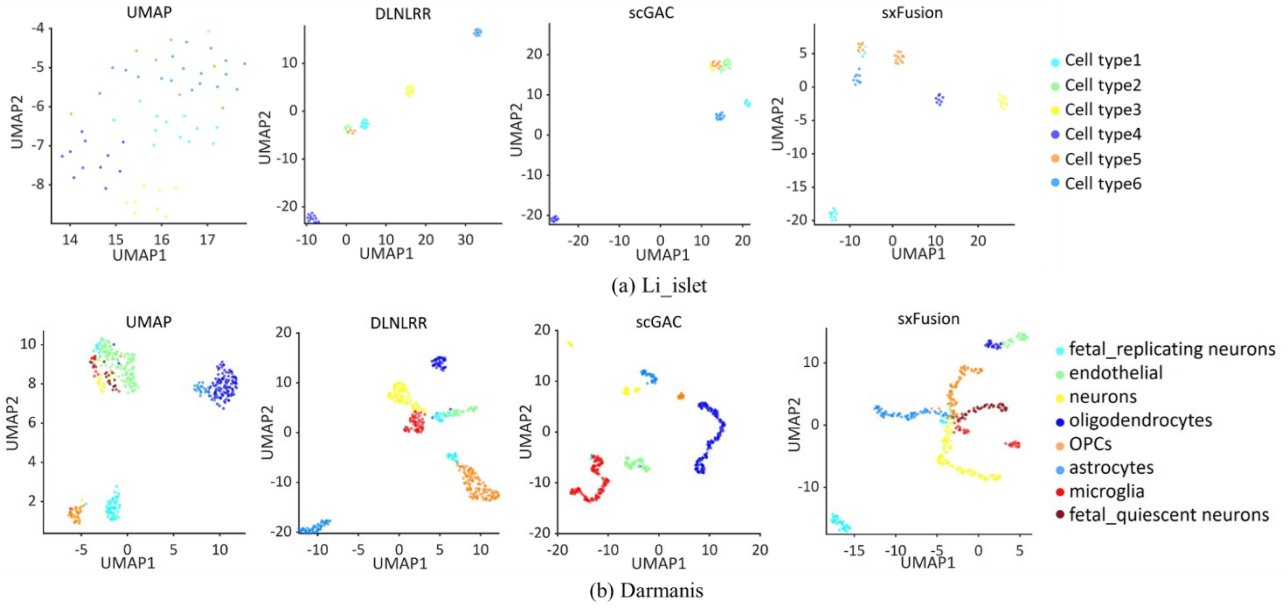


Fig. 5 UMAP visualization of clustering result on Li_islet and Darmanis dataset. Colors represent different cell types, showing clear separation achieved by sxFusion (right) compared to baseline methods.

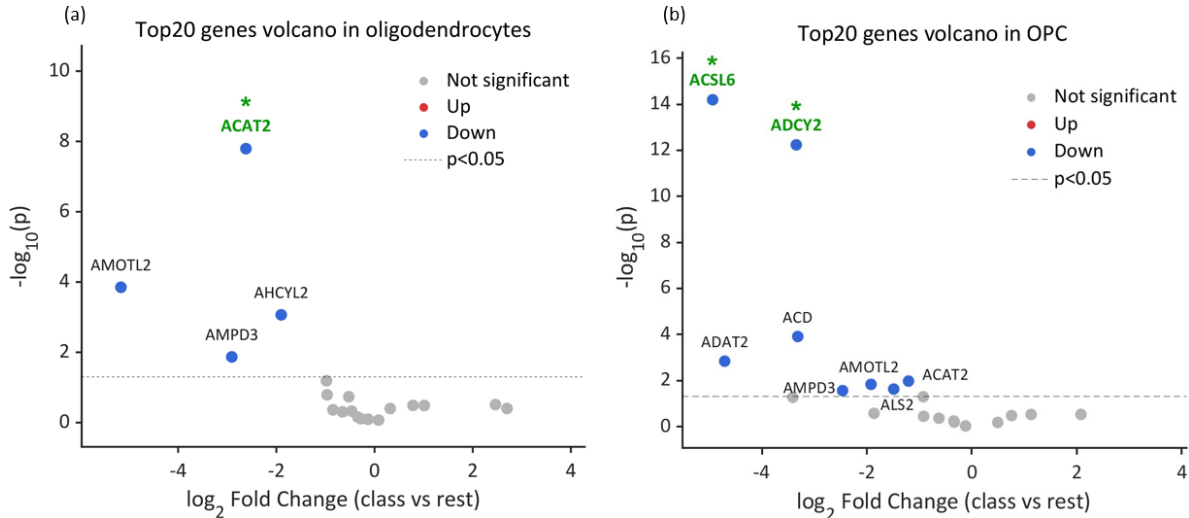


Fig. 6 Volcano plots display one-vs-rest differential expression for sxFusion-derived top 20 global marker genes in each brain cell type. red: upregulated, blue: downregulated, gray: not significant; green asterisk: Bonferroni is still significant after correction.

degradation, especially under challenging conditions. This highlights the synergistic benefit of jointly learning structure-aware representations and clustering assignments in single-cell transcriptomic analysis.

D. Visualization

To assess the practical utility of sxFusion, we used UMAP [20] to visualize sample distributions for DLNLRR, scGAC, and sxFusion on islet and Darmanis datasets, as shown in Fig. 5. The following observations are based on these visualizations.

For the islet dataset, all methods produce clusters that are relatively compact and comparable to the raw data structure. However, a clear difference emerges on the Darmanis dataset. DLNLRR and scGAC tend to underestimate the number of clusters and fail to recover the main cell types. In contrast,

sxFusion identifies a cluster structure that aligns more closely with the annotated cell populations. This advantage may stem from sxFusion’s dimension-wise gating, which adaptively balances graph topology and attribute signals, as well as its joint low-rank and NMF optimization, which encourages cluster-aware embeddings during training.

E. sxFusion-Based Differential Gene Expression Analysis

To identify and validate robust marker genes that distinguish cell types after clustering the Darmanis dataset with sxFusion, a global importance score was computed for each gene using the genes-by-clusters indicator matrix, and the top 20 genes were selected as global markers [6], as shown in Fig. 6. One-vs-rest differential testing for each cluster was then performed, and the results were visualized with volcano plots. Many markers, such as ACSL6 and ACAT2, showed significant and cell type–

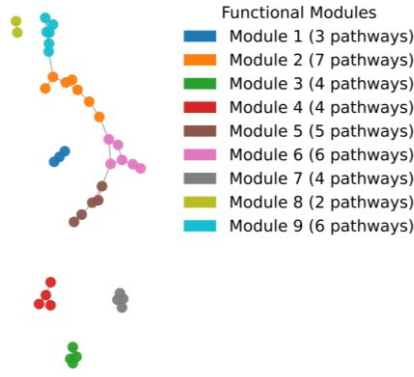


Fig. 7 Pathway network derived from pathways enriched in sxFusion clusters using the Darmanis dataset. Each node represents an enriched pathway, colored by its functional module as identified by Louvain clustering. Edges indicate strong co-activation similarity between pathways. Module labels, such as "Module 9 (6 pathways)" indicate the number of pathways within each module. The network structure reveals both a backbone of interconnected modules and several isolated modules, highlighting the balance between coordinated and specialized biological programs among clustered cell types.

specific expression, confirming that sxFusion identifies robust and functionally meaningful clusters enriched for pathways like lipid metabolism and cAMP signaling.

Based on the clusters identified by sxFusion, we performed differential expression analysis to identify cluster-specific marker genes, followed by pathway enrichment analysis of these marker sets, as shown in Fig. 7. To gain systems-level insight, we constructed a pathway network by connecting pathway pairs whose enrichment score profiles (NES) across clusters exhibited strong Pearson correlation ($|r| > 0.7$). Louvain community detection was applied to partition the network into functional modules [21]. The resulting network revealed both tightly connected modules and a backbone-like chain of pathways, reflecting functional specialization and coordination among cell populations. Functional annotation indicated that the core pathways within each module are enriched for biological processes associated with cluster-specific marker genes identified in differential analyses. Notably, the observed overlap between module-enriched pathways and marker gene functions was substantially higher than expected by chance, supporting the biological relevance and interpretability of the identified modules. These results demonstrate that sxFusion clustering, combined with pathway network analysis, links molecular signatures to higher-order functional organization, providing a mechanistic framework for understanding cellular heterogeneity.

IV. CONCLUSION

In this work, we present sxFusion, a novel single-cell clustering method that aims to improve clustering performance by introducing three innovations: (1) gating-based graph-attribute feature fusion mechanism; (2) joint optimization of LRR and clustering via orthogonal NMF; (3) and direct cell type assignment without post-processing. In a comprehensive evaluation across 11 diverse scRNA-seq datasets, sxFusion achieves competitive and consistent clustering performance, with NMI and ARI values reaching 0.980 and 0.988, respectively, even under extreme sparsity (88.76% dropout). On average, it improves clustering performance by 19.3% compared with selected state-of-the-art methods. The approach

identifies biologically meaningful subpopulations, as indicated by marker gene and pathway enrichment analyses. Future directions include extending sxFusion for multi-omics integration, temporal and spatial analysis, and application to large-scale or clinical single-cell datasets, further advancing our understanding of cellular heterogeneity in health and disease.

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